BHANNAK ABHILASH

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Education

B V Raju Institute of Technology Narsapur, Medak
Bachelor of Technology in Electronics and Communication Engineering (CGPA: 8.82) *Oct 2022 – May 2026*

SV Junior College Sangareddy
Intermediate in Mathematics, Physics, and Chemistry (Marks: 974) *Aug 2020–May 2022*

Sai Krishnaveni High School Jogipet
Secondary School Certificate (CGPA: 10) *April 2020*

Experience

Python Trainee Intern
Trysol Global Services Pvt. Ltd. *Sep 2025 – Jan 2026*

Project: Skill Synch – Skill-Based Learning Platform
Technologies: Python, SQL

- Contributed to backend development for learner and mentor management modules using Django.
- Worked on database integration and CRUD operations using PostgreSQL.
- Supported authentication and session management features for secure user access.
- Coordinated with frontend developers for API integration and response handling.
- Participated in testing, debugging, and resolving functional issues in platform modules

Competitions and Hackathons

Digital Twin for Battery Management System - L&T Techgium 2025 - Runner Up Feb 2025 – Apr 2025
Technologies: Python, LSTM, Machine Learning, Battery Management System (BMS), Data Analysis

- Built a Digital Twin to eliminate sensor inefficiencies and improve battery health monitoring.
- Achieved 95% accuracy between Digital Twin and hardware data using key battery parameters.
- Trained an LSTM model on custom dataset for accurate SOC prediction.
- Created an Enhanced Mode that fuses real and simulated data, delivering more reliable BMS outputs.

Smart India Hachathon (SIH) - Wearable Safety Device for Women Aug 2024

- Discreet panic button triggers GPS, audio, video capture for emergencies.
- Automatic alerts for faster emergency response time.
- Low-power mode ensures battery efficiency until device activation.

Academic Projects

Augmentative and Alternative Communication Jun 2024 – Aug 2024
Technologies: Python, IoT, and FPGA

- Developed an Augmentative and Alternative Communication (AAC) system to assist users with speech or motor impairments in daily communication.
- Ensured multi-input accessibility (keyboard, touch, and mouse) and optimized the system for real-time response and ease of use.

Multi Metric Analysis of Power Efficient Digital Circuits Apr 2025– March 2026
Technologies: Verilog, CMOS Logic, Adiabatic Logic, Python, CSV Processing, Data Analysis

- Designed and implemented adiabatic logic-based digital circuits in Verilog to enhance power efficiency over conventional CMOS designs.
- Developed and simulated digital circuits in Vivado, comparing power consumption between adiabatic and traditional approaches.
- Demonstrated significant power savings through energy recovery, validating results using Vivado's simulation and analysis tools.
- Export of simulation result in CSV format allows for simple handling of the data and integration with the python based dashboard for analysis.

Technical Skills

Programming Languages: Python, C, SQL, Verilog, System Verilog
Tools: VS code, Git, Microwind, Arduino, Matlab, Vivado
Frameworks & Technologies: Gen Ai, Prompt Engineering
Hardware: FPGA: PYNQ board

Certifications

- Smart interviews: Smart Coder
- LetsUpgrade : Python Bootcamp
- NPTEL: System Design Through Verilog
- Udemy: System Verilog (SV) language